

EMCH 260

SOLID MECHANICS

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Torsion

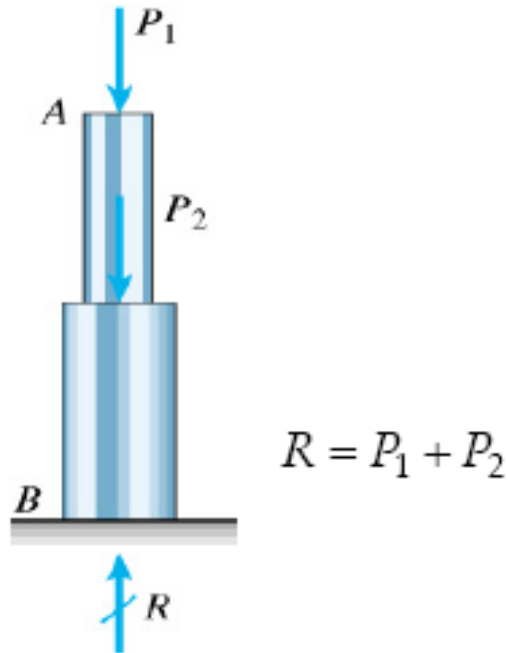
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Chapter 5.3

- ✓ Understand the statically indeterminate torque-loaded members
- ✓ Deal with compatibility conditions for torsion
- ✓ Solve the statically indeterminate torque-loaded problems

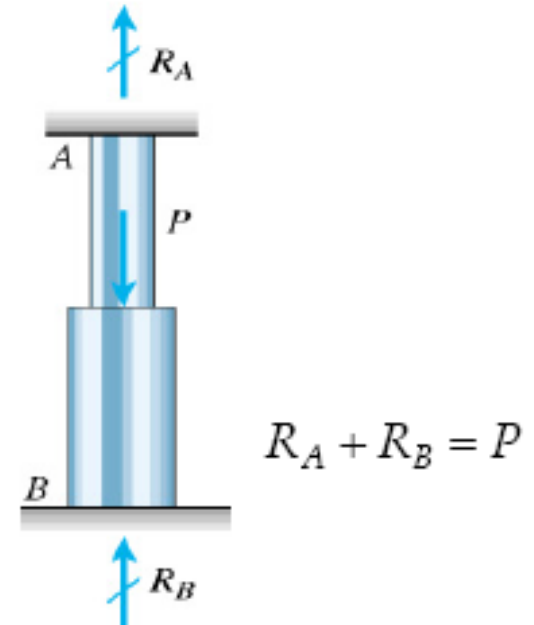
REVIEW: DETERMINATE VS. INDETERMINATE AXIALLY LOADED MEMBER

Statically Indeterminate Structures



Statically Determinate Structures

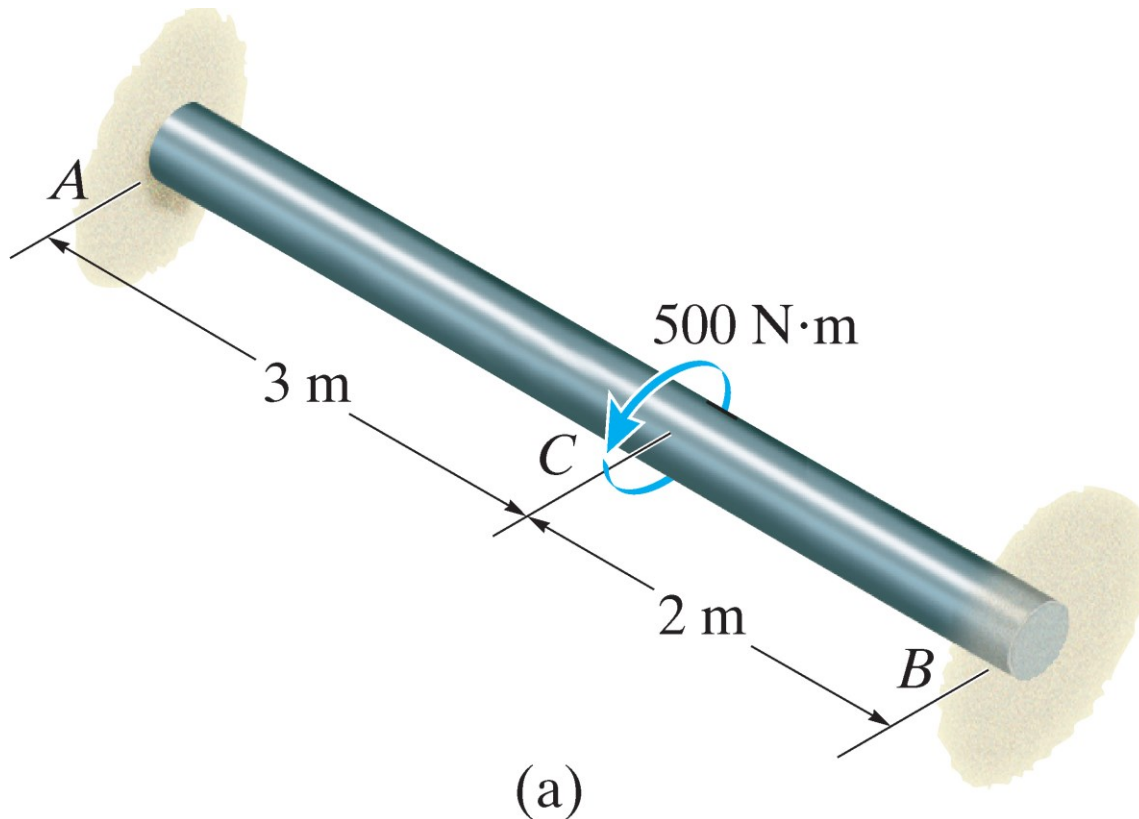
Reactions and internal forces can be determined solely from free-body diagrams and *equations of equilibrium* without knowing the properties of the materials.



Statically Indeterminate Structures

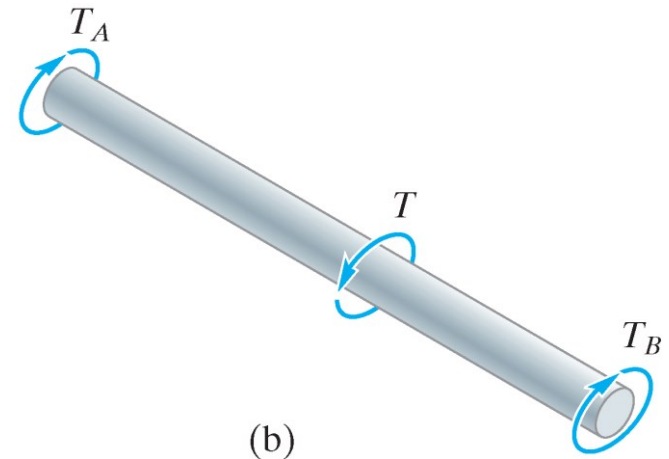
In addition to the *equilibrium consideration*, equations pertaining to the displacement of the structure, called *equations of compatibility*, and the *relations between forces and displacements* are usually needed.

STATICALLY INDETERMINATE TORQUE-LOADED MEMBER



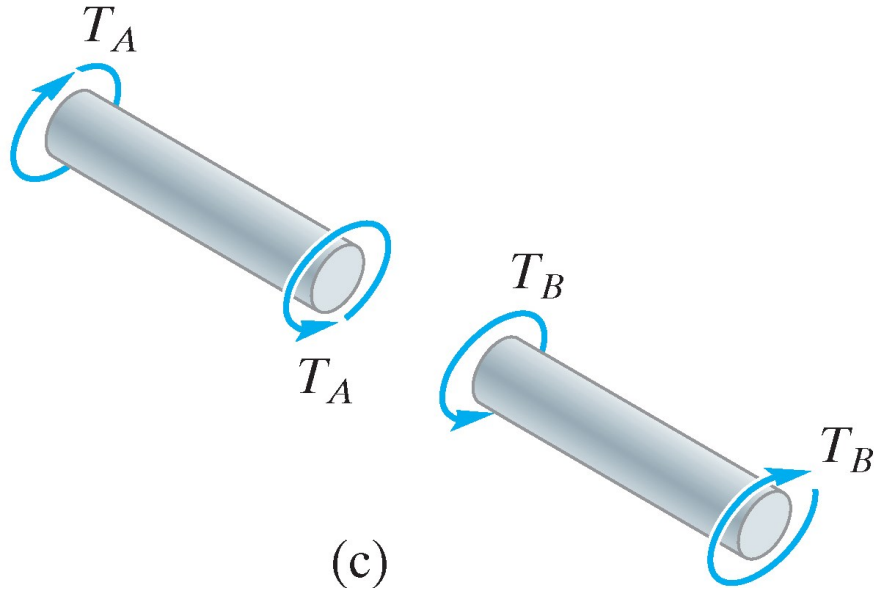
$$\sum M = 0$$

$$\rightarrow 500 - T_A - T_B = 0 \quad (1)$$



- The moment equation of equilibrium is not adequate to determine the unknown torques acting on the shaft

COMPATIBILITY CONDITION



- The necessary compatibility condition requires the angle of twist of one end of the shaft with respect to the other end to be equal to zero, since both end supports are fixed.

$$\phi_{A/B} = 0$$

$$\rightarrow \frac{T_A L_{AC}}{JG} - \frac{T_B L_{BC}}{JG} = 0 \quad (2)$$

Combining Eqs. (1) and (2), we obtain

$$T_A = 200 \text{ N} \cdot \text{m}, T_B = 300 \text{ N} \cdot \text{m}$$

PROCEDURE FOR ANALYSIS

Step 1: Equilibrium (exactly same as what we did before)

- Draw a free-body diagram of the shaft to identify torques acting on it
- If unknown reactions on free-body diagram greater than number of moment equilibrium equations, then problem is statically indeterminate
- Write the equations of moment equilibrium for the shaft

Step 2: Compatibility

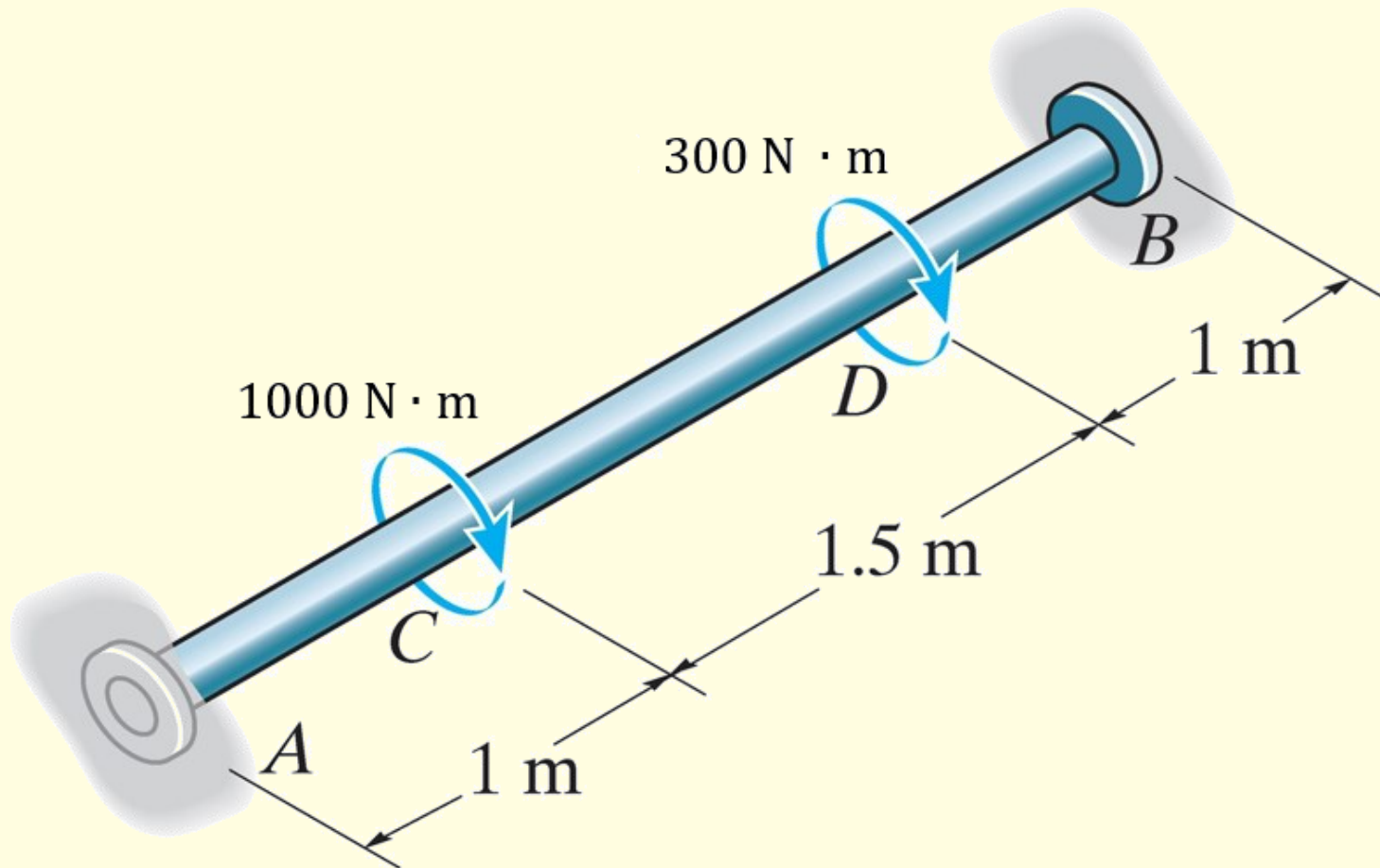
- Express compatibility conditions in terms of twist angles caused by torques

Step 3: Load-Displacement

- Use load-displacement relations (such as $\phi = TL/(JG)$) to relate unknown twist angles to reactions
- Solve the equations for the unknown reactive torques. If any of the magnitudes is negative, it means the torque acts in opposite direction of that indicated on free-body diagram.

IN-CLASS EXAMPLE

The steel shaft is fixed at its two ends A and B . If it is subjected to two external torques ($1000 \text{ N}\cdot\text{m}$ and $300 \text{ N}\cdot\text{m}$ at point C and D , respectively) as shown in the figure, determine the support reactions at two ends A and B .



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 1: Equilibrium

Assuming the resultant reactions at the two ends A and B as shown in the figure (right hand rule gives positive x -direction for both reactions), for the entire shaft, the Free-Body-Diagram (moment equation of equilibrium) gives:

$$\sum M = 0 \rightarrow$$

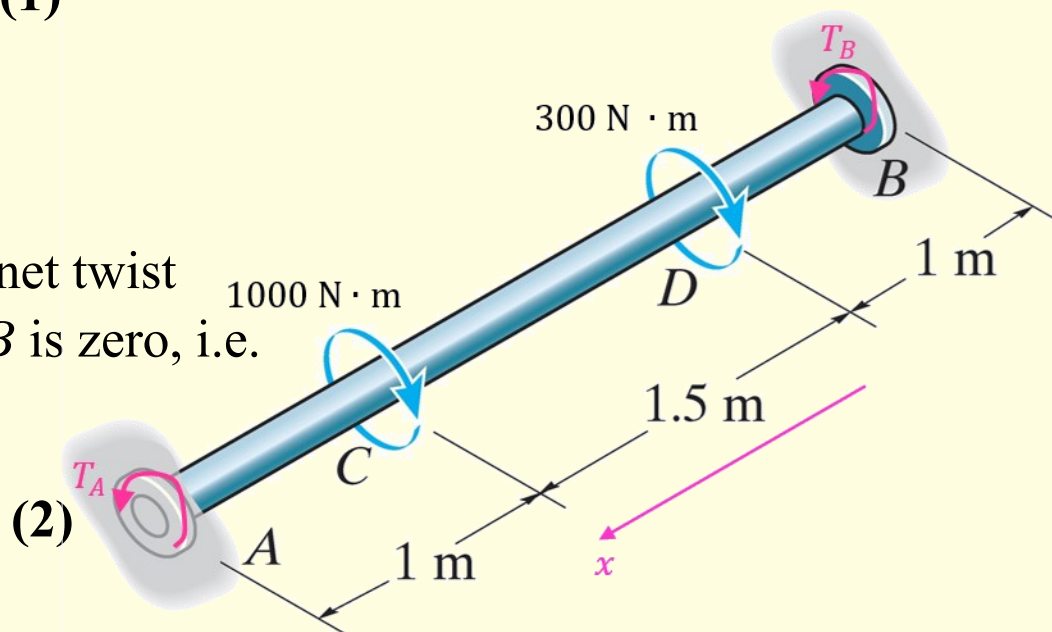
$$T_A + T_B = 1000 + 300 = 1300 \quad (1)$$

Step 2: Compatibility

Since both ends are fixed, the total net twist angle of end A with respect to end B is zero, i.e.

$$\phi_{AB} = 0 \rightarrow$$

$$\phi_{AC} + \phi_{CD} + \phi_{DB} = 0$$



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 3: Load-Displacement

Using the sign convention, the angle of twist for each segment of the bar can be obtained by the torque, shear modulus of elasticity, and the polar moment of inertia:

$$\phi_{AC} = + \frac{T_A L_{AC}}{JG}$$

$$\phi_{CD} = + \frac{(T_A - 1000)L_{CD}}{JG}$$

$$\phi_{DB} = - \frac{T_B L_{DB}}{JG}$$

Substituting the above equations into Eq. (2), we get

$$\frac{T_A L_{AC}}{JG} + \frac{(T_A - 1000)L_{CD}}{JG} - \frac{T_B L_{DB}}{JG} = 0$$

$$\rightarrow T_B = 2.5T_A - 1500 \quad (3)$$

IN-CLASS EXAMPLE (cont'd)

Solutions

Combining Eqs. (1) and (3), we obtain

$$T_A = 800 \text{ N} \cdot \text{m} \quad (\text{Ans.})$$

$$T_B = 500 \text{ N} \cdot \text{m} \quad (\text{Ans.})$$

Note: Both reaction moments are positive, indicating that our presumed direction is correct.

RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #5-6 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**
- **Optional Extra Exercise 2 on Blackboard, problem #4 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 222 – 225**
 - **In particular, Examples 5.9 – 5.10 (pp. 224 – 225)**